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Thermodynamic Modeling of the Al-Li-Zr Ternary System

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Abstract Using CALPHAD method, the Al-Zr, Al-Li and Al-Li-Zr systems have been reassessed based on the latest experimental phase relations from literatures and the first-principle calculation of the formation enthalpy for ternary compound T1($AlLi_2Zr$) in this work. The excess Gibbs energies of solution phases, including liquid, bcc, fcc and hcp, were expressed by the Redlich–Kister polynomial. The stoichiometric compounds, i.e. Al_4Zr_5, Al_3Zr_5, $AlZr_2$, $AlZr_3$, Al_2Li_3, Al_4Li_9 and T1($AlLi_2Zr$), were modeled as stoichiometric model and the non-stoichiometric compounds, i.e. Al_3Zr, Al_2Zr, Al_3Zr_2, AlZr, Al_3Zr_4, Al_2Zr_3, $AlLi_2$ and T2($AlLi_{x+y}Zr_{5-x}$), were described with different sublattice models. Finally, a set of reasonable thermodynamic parameters for Al-Li-Zr ternary system have been obtained, which notes a clear improvement on the self-consistency. The calculated isothermal section at 470 K of Al-Li-Zr system was in reasonable agreement with the experimental one.

Keywords Al-Li-Zr · Al-Zr · CALPHAD · phase diagram

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1 Introduction

Aluminum and its alloys are known as the materials used for aviation, aerospace, automotive and etc. Zirconium and Lithium are important alloy elements for aluminum. Mechanical properties of aluminum alloy are improved by adding Zr and Li.[1,2] In Al-Li-Zr ternary system, Al-Zr covalent bond in Al_3Zr is greater than the Al-Li covalent bond in Al_3Li, leading to Al_3Zr using as an inhomogeneous nucleus for Al_3Li, which improve the strength and toughness of alloy.[3] Studied by classical nucleation theory, Al_3Zr is a nucleation substrate for Al_3Li.[4] Furthermore, Wang et al.[5] reported bimodal-grained Al-Li/Al-Li-Zr laminated composite owns high strength and ductility.

It is realized that the calculation of phase diagrams (CALPHAD) approach is a useful tool to establish thermodynamic databases which bases for the predictions of thermodynamic properties, phase diagrams, DICTRA[6] and phase-field simulations.[7]

According to the latest report,[8] the liquidus and solidus between 20 and 94.6 at.% Zr at 1573–2133 K of Al-Zr system have been measured. Furthermore, a new phase $AlLi_2$ has been discovered in Al-Li binary system[9] and two new ternary compounds have been found in Al-Li-Zr ternary system.[10] However, the experimental information mentioned above was not considered by recent assessments.[11–17] In order to describe these systems more reasonably, the thermodynamic reassessment of Al-Li-Zr ternary system is necessary. In this work, Al-Zr and Al-Li binary systems have been reassessed according to the latest experimental phase diagrams.[8,9] Then, combining the earlier thermodynamic assessment for the binary Li-Zr system, the thermodynamic description of the Al-Li-Zr ternary system has been carried out based on the experimental isothermal section at 470 K of Al-Li-Zr ternary

system reported by Zatorska.[10] The crystallographic parameters for the binary and ternary compounds involved in the present work are listed in Table 1.

2 Literature Data

2.1 Al-Zr Binary System

The invariant equilibrium data of Al-Zr binary phase diagram was reported by several researchers.[8,19–28] All these results are summarized in Table 2. The first systematic and comprehensive investigation of Al-Zr phase diagram was performed by McPherson et al.[19] using metallography, x-ray diffraction, incipient melting and thermal analysis. The partial invariant reactions of Al-Zr phase diagram were measured by Pötzschke et al.,[20] Peruzzi et al.[21] and Fink et al.[22] Subsequently, the Al-rich part of the Al-Zr phase diagram, including liquidus and invariant reaction information, has been investigated with means of metallography observation and differential thermal analysis (DTA) by Janghorban et al.[23] Wang et al.[24] and Khvan et al.[25] reported that the reaction between fcc(Al) and Al_3Zr is of eutectic nature, which identify with the result measured by Janghorban et al.[23]

The liquidus in Al-rich part was measured by Fink et al.,[22] Chiotti et al.[26] and Dezellus et al.[27] from 939 to 1072 K. They arrived in similar results. In addition, Janghorban et al.[23] determined the liquidus between 0 and 20 at.% Zr above 1072 K. The solubility of Al in hcp(Zr) was determined by McPherson et al.[19] and Tiwari et al.[28]

Recently, Samanta et al.[8] measured the invariant equilibria between 1573 and 2133 K of Al-Zr binary system. Meanwhile, with the spot-technique, the liquidus and solidus of Al-Zr binary phase diagram between 20 and 94.6 at.% Zr at 1573–2133 K were measured by Samanta et al.[8] at the first time.

Al-Zr binary system contains ten stable intermetallic compounds. The crystal structures have been researched by many researchers[28–33] and the results are listed in Table 1. The formation enthalpies of the compounds in Al-Zr binary system have been investigated with both experimental method[34–37] and the first-principle method.[38–41] The partial and integral mixing enthalpies of liquid Al-Zr alloys were measured by Esin et al.[42] at 1970 K, Sudavtsova et al.[43] at 1693 K, Witusiewicz et al.[44] at 2045 K, Sudavtsova et al.[45] at 1790 K and Gajavalli et al.[46] at 1173 K. The results mentioned above were given different weight in the present work to optimize the parameter of liquid phase.

Thermodynamic assessments of Al-Zr binary system were performed by Saunders and Rivlin,[11] Saunders,[12] Murray et al.,[13] Wang et al.,[14] Fischer et al.[15] and Tamim et al.,[16] respectively. According to the phase diagram calculated by Fischer et al.[15], the Al_2Zr_3 phase did not exist at 470 K, which was inconsistent with the

Table 1 Crystallographic data of the phases in the Al-Zr,[28–33] Al-Li[9,53,57,58] and Al-Li-Zr[10] system

Phase	Space group	Prototype	Pearson symbol	Lattice parameter, Å		
				a	b	c
fcc(Al)	*Fm* $\bar{3}$*m*	Cu	*cF4*	4.0496		
bcc(Li)	*Im3m*	W	*cI2*	3.5093		
hcp(Zr)	*P6*$_3$*/mmc*	Mg	*hP2*	3.2317		5.1475
Al_3Zr	*14/mmm*	Al_3Zr	*tI16*	3.998		17.28
Al_2Zr	*P6*$_3$*/mmc*	$MgZn_2$	*hP12*	5.282		8.7482
AlZr	*Cmcm*	BCr	*oC18*	3.353	10.866	4.266
Al_4Zr_5	*P6*$_3$*/mcm*	Ga_4Ti_5	*hP18*	8.447		5.810
Al_3Zr_4	*p* $\bar{6}$	Al_3Zr_4	*hP7*	5.433		5.390
Al_3Zr_2	*Fdd2*	Al_3Zr_2	*oF40*	9.599	13.879	5.572
Al_2Zr_3	*P4*$_2$*/mm*	Al_2Zr_3	*tP20*	7.630		6.998
Al_3Zr_5	*14/mcm*	Si_3W_5	*tI32*	11.049		5.396
$AlZr_2$	*P6*$_3$*/mmc*	Ni_2In	*hP6*	4.894		5.928
$AlZr_3$	*Pm* $\bar{3}$*m*	$AlCu_3$	*cP4*	4.3917		
AlLi	*Fd* $\bar{3}$*m*	NaTi	*cF16*	6.36		
Al_2Li_3	*R* $\bar{3}$*m*	Bi_2Te_3	*hR15*	4.508		14.259
Al_4Li_9	*C2/m*	Al_4Li_9	*mC26*	19.151	5.429	4.499
$AlLi_2$	*Cmcm*	$GaLi_2$		4.658	9.767	4.490
$AlLi_2Zr$	*F* $\bar{4}$*3m*	$TiHg_2Cu$		6.633		
$Al_3Li_{x+y}Zr_{5-x}$	*P6* $\bar{3}$*/mcm*	Ga_4Ti_5		8.1336–8.1757		5.7029–5.6909

Table 2 Invariant reactions as compared to literature information in Al-Zr binary system

Invariant reaction	Temperature (K)	Composition of the coexisting phase at.% Zr			References
$L \leftrightarrow bcc(Zr) + Al_3Zr_5$	1623	70.5	74.1	62.5	[19]
	1629	70.8	75.8	62.5	This work
$bcc(Zr) + Al_3Zr_5 \leftrightarrow AlZr_2$	1488	80.1	62.5	66.7	[21]
	1490	80.4	62.5	66.7	This work
$bcc(Zr) + AlZr_2 \leftrightarrow AlZr_3$	1292	88	66.7	75	[21]
	1296	88.1	66.7	75	This work
$bcc(Zr) + AlZr_3 \leftrightarrow hcp(Zr)$	1183	93.2	75	91.7	[21]
	1185	93.0	75	91.2	This work
$L + Al_2Zr_3 \leftrightarrow Al_3Zr_5$	1673	65	60	62.5	[19]
	1667	68.4	60	62.5	This work
$Al_3Zr_5 \leftrightarrow AlZr_2 + Al_2Zr_3$	1275	62.5	66.7	60	[19]
	1273	62.5	66.7	60	[20]
	1277	62.5	66.7	60	This work
$L + Al_4Zr_5 \leftrightarrow Al_2Zr_3$	1753	61	56.6	60	[19]
	1754		56.6	60	[8]
	1748	62.7	56.6	60	This work
$Al_2Zr_3 + Al_4Zr_5 \leftrightarrow Al_3Zr_4$	1303	60	56.6	57.1	[20]
	1298	60	56.6	57.1	This work
$Al_4Zr_5 \leftrightarrow Al_3Zr_4 + AlZr$	1273	56.6	57.1	50	[20]
	1277	56.6	57.1	50	This work
$L \leftrightarrow Al_4Zr_5 + Al_3Zr_2$	1758	50	56.6	40	[19]
	1753	50	56.6	40	[8]
	1756	48.9	56.6	40	This work
$Al_4Zr_5 + Al_3Zr_2 \leftrightarrow AlZr$	1548	56.6	40	50	[19]
	1543	56.6	40	50	This work
$L + Al_2Zr \leftrightarrow Al_3Zr_2$	1868	41	33.3	40	[19]
	1845	42	33.3	40	[23]
	1846	42	33.3	40	[8]
	1845	42.1	33.3	40	This work
$L + Al_2Zr \leftrightarrow Al_3Zr$	1852	24	33.3	25	[23]
	1855	23.8	33.3	25	This work
$L \leftrightarrow fcc(Al) + Al_3Zr$	929	0.72	0.08	25	[23]
	933	0.029	0.015	25	[25]
	933	0.036	0.015	25	This work
$L \leftrightarrow Al_2Zr$	1918	33.3	33.3		[19]
	1897	33.3	33.3		[23]
	1922	33.3	33.3		This work
$L \leftrightarrow Al_4Zr_5$	1823	56.6	56.6		[19]
	1825	56.6	56.6		[8]
	1820	56.6	56.6		This work

experimental isothermal section of the Al-Li-Zr ternary system.[10] Meanwhile, the phase diagram calculated by Tamim et al.[16] did not agree well with the experimental data about solid solubility of Al in hcp(Zr). Moreover, all of the assessments[11–16] of Al-Zr system mentioned above didn't consider the experimental information about the liquidus and solidus between 20 and 94.6 at.% Zr which was reported later.[8] Therefore, the Al-Zr binary system will be thermodynamically reassessed and the latest experimental data will be adopted in this work.

2.2 Al-Li Binary System

The Al-Li binary system has been investigated by a number of researchers on the phase equilibria, crystal structures and thermodynamic information. The phase equilibria information was measured by some investigators.[47–56] According to the early research, Al-Li system was characterized by three stable intermediate phases: AlLi (β), Al_2Li_3 and Al_4Li_9. As pointed by Myles et al.[50], Al_2Li_3 and Al_4Li_9 phase have small homogeneity range, and AlLi phase has an obvious homogeneity range.

More recently, Puhakainen et al.[9] discovered a new phase in Al-Li binary system, i.e. orthorhombic $AlLi_2$, whose formula is $Al_{1.08}Li_{1.92}$. $AlLi_2$ is formed by a peritectic reaction of Al_2Li_3 with liquid at about 608 K. According to the report,[9] $AlLi_2$ may have a small homogeneity range $Al_{1+x}Li_{2-x}$, and the value of 'x' is approximately equal to 0.08. This latest experimental information[9] will be used in this work. The information of the invariant reactions is listed in Table 3.

The crystal structures of Al_2Li_3, Al_4Li_9 and AlLi were determined by Tebbe et al.,[57] Hansen et al.[58] and Kishio et al.[53] respectively. Finally, the crystal structures data of Al-Li binary system is listed in Table 1.

With ab-initio method, the formation enthalpies of the Al-Li compounds were determined by Christensen et al.[59], Guo et al.[60], Arya et al.[61] and Sluiter et al.[62] respectively. Afterwards, the formation enthalpy of AlLi was determined from EMF data by Yao et al.[63]. Recently, the formation enthalpies of AlLi, Al_2Li_3 and Al_4Li_9 were measured using the two calorimetric techniques by Gasior et al.[64] The above-mentioned information on formation enthalpies of Al-Li compounds is summarized in Table 4.

The Al-Li binary system has been assessed by many investigators. Recently, the thermodynamic parameters of the Al-Li phase diagram were assessed by Hallstedt and Kim,[17] which can reproduce experimental information.[47–56] satisfactorily However, the calculated phase diagram[17] didn't include $AlLi_2$ which was discovered by Puhakainen et al.[9] later. Therefore, considering the new phase $AlLi_2$,[9] Al-Li binary system has been reassessed in this work basing on the calculated Al-Li phase diagram[17] (Table 3).

2.3 Li-Zr Binary System

The Li-Zr binary phase diagram has been studied by some researchers. This system is essentially completely immiscible in both the solid and the liquid phases. Bychkov et al.[65] measured the solubility of Zr in liquid Li varies from approximately 0.0001 at.% Zr at 973 K to 0.033 at.% Zr at 1473 K. The Li-Zr binary diagram was assessed by Bale.[66] Later, the thermodynamic parameters were determined by Saunders[12] and Long et al.[8] using CALPHAD method, respectively. The thermodynamic parameters reported by Long et al.[8] have been successfully used to build the thermodynamic databases for Li-Si-Zr ternary system. Therefore, in the present work, most of the thermodynamic parameters in Ref. [8] are adopted directly and presented in Table 5.

2.4 Al-Li-Zr Ternary System

The isothermal sections of Al-Li-Zr ternary system at 373, 573 and 773 K in the Al-rich corner have been predicted by Saunders.[12] The liquidus surface in the Al-rich corner also has been calculated by Saunders.[12] However, no excess ternary parameters have been used in their work because of lacking experimental information.

Subsequently, the isothermal section of the Al-Li-Zr system at 470 K was built using x-ray structural analysis by Zatorska et al.[10] Two ternary compounds, i.e. T1 ($AlLi_2Zr$) and T2 ($AlLi_{x+y}Zr_{5-x}$), have been found and the crystal structures have been determined. According to their results,[10] the T1 phase is in the $TiHg_2Cu$ structure type and the T2 phase crystallizes in the Ga_4Ti_5 structure type. These experimental results[10] have been considered in the present work.

3 Thermodynamic Modeling

3.1 Unary Phase

The equation for the molar Gibbs energy of pure element i (i = Al, Li or Zr) relative to the standard element reference is as follow:

$$G_i(T) - H_i^{\mathrm{SER}} = A + \mathrm{BT} + \mathrm{ClnT} + \mathrm{DT}^2 + \mathrm{ET}^{-1} + \mathrm{FT}^3 + \mathrm{JT}^7 + \mathrm{HT}^{-9} \quad \text{(Eq 1)}$$

where H_i^{SER} is the molar enthalpy of the element i at 298.15 K and 1 bar in its standard state, and T is the absolute temperature. The coefficients A through J for pure elements are taken from the SGTE data compilation of Dinsale.[67]

3.2 Solution Phase

The solution phase, namely liquid, fcc(Al), bcc(Zr) and hcp(Zr) are described by an ordinary sub-regular solution model. The Gibbs energy of the solution phases is given by the following equation:

Table 3 Invariant reactions as compared to literature information in Al-Li binary system

Invariant reaction	Temperature	Composition of the coexisting phase			References
	K	at.% Li			
L ↔ AlLi	965				[50]
	970	49	49		[51]
	975	49.7	49.7		[17]
L ↔ fcc(Al) + AlLi	863	23.4	12.4		[47]
	873	30	18		[48]
	871	26			[49]
	873				[50]
	875	25.2	13.8	47.5	[51]
	876	23.8	13.1	45.6	[17]
	876	23.8	13.1	45.6	This work
AlLi + L ↔ Al_2Li_3	794	56	78	66.7	[48]
	793		70		[49]
	793			60	[50]
	792	58	75.6	66.7	[51]
	793	56.7	74.5	60	[17]
	793	56.7	74.5	60	This work
Al_2Li_3 + L ↔ Al_4Li_9	608	60		69.2	[50]
	620	60	920.8	69.2	[52]
	610	60	91.6		[17]
	610	60	91.6		This work
L + Al_4Li_9 ↔ bcc(Li)	453		69.2		[50]
	453.6	99.4	69.2		[52]
	452.3	99.32	69.2	99.57	[17]
	452.3	99.32	69.2	99.57	This work
Al_2Li_3 + L ↔ $AlLi_2$	608	60	89.5	64.2	[9]
	608	60	91.3	65.1	This work
L + $AlLi_2$ ↔ Al_4Li_9	543	95.0	66.3	69.2	[9]
	545	96.3	65.8	69.2	This work

$$G_m^{\Phi} = \sum_i x_i^{\Phi} {}^0G_i^{\Phi} + \mathrm{RT}\left(\sum_i x_i^{\Phi} \ln x_i^{\Phi}\right) + {}^{\mathrm{ex}}G_m^{\Phi} \qquad \text{(Eq 2)}$$

where R is the gas constant, T is the absolute temperature, and Φ is the indicator of the liquid or solid solution phase. x_i^{Φ} and ${}^0G_i^{\Phi}$ are the mole fraction and the molar Gibbs energy of the species i (i = Al, Li or Zr). ${}^{\mathrm{ex}}G_m^{\Phi}$ is the excess Gibbs energy, which can be expressed in Redlich–Kister polynomial[68] as follow:

$$\begin{aligned} {}^{\mathrm{ex}}G_m^{\Phi} = & \sum_i \sum_{j>i} x_i^{\Phi} x_j^{\Phi} \sum_v {}^vL_{i,j}^{\Phi}\left(x_i^{\Phi} - x_j^{\Phi}\right)^v \\ & + \sum_i \sum_{j>i} \sum_{k>j} x_i^{\Phi} x_j^{\Phi} x_k^{\Phi}\left(x_i {}^0L_{i,j,k}^{\Phi} + x_j {}^1L_{i,j,k}^{\Phi} + x_k {}^2L_{i,j,k}^{\Phi}\right) \end{aligned} \qquad \text{(Eq 3)}$$

where $L_{i,j}^{\Phi}$ and $L_{i,j,k}^{\Phi}$ are the binary and ternary interaction parameters, respectively. They can be described in empirical formula:

$$L^{\Phi} = A + BT \qquad \text{(Eq 4)}$$

where A and B are lattice stability parameters to be evaluated.

3.3 Intermetallic Phase

The intermetallic compounds in the Al-Li-Zr ternary system can be divided into two categories during the optimization: stoichiometric phases, i.e. Al_4Zr_5, Al_3Zr_5, $AlZr_2$, $AlZr_3$, Al_2Li_3, Al_4Li_9 and T1, phases with solubility ranges, i.e. Al_3Zr, Al_2Zr, Al_3Zr_2, AlZr, Al_3Zr_4, Al_2Zr_3, $AlLi_2$, and T2.

The stoichiometric phases are described as stoichiometric model because of the limited solubility. The Gibbs energies given by the following equation:

$$G_m^{\mathrm{Al}_a\mathrm{Li}_b\mathrm{Zr}_c} = \frac{a}{a+b+c}{}^0G_A^{\mathrm{HSER}} + \frac{b}{a+b+c}{}^0G_B^{\mathrm{HSER}} + \frac{c}{a+b+c}{}^0G_C^{\mathrm{HSER}} + A + BT \quad \text{(Eq 5)}$$

where the coefficients A and B are parameters to be optimized. The value of b or c can be zero to present binary compounds in Al-Zr or Al-Li binary systems.

Different sublattice models are used to describe the phases with significant homogeneity ranges in the Al-Li and Al-Li-Zr systems. The two-sublattice model, i.e. $(\mathrm{Al,Li})_a\mathrm{A}_b$, was used to describe the phase with homogeneity ranges, such as Al_3Zr, Al_2Zr, Al_3Zr_2, AlZr, Al_3Zr_4, Al_2Zr_3, and $AlLi_2$. The Gibbs energy per mole-formula can be described using an equation as follow:

$$ {}^0G_m^{(\mathrm{Al,Li})_a\mathrm{A_b}} = y_{Al}^{(1)}G_{\mathrm{Al}:A} + y_{\mathrm{Li}}^{(1)}G_{\mathrm{Li}:A} + \frac{a}{a+b}RT\left(y_{\mathrm{Al}}^{(1)}\ln y_{\mathrm{Al}}^{(1)} + y_{\mathrm{Li}}^{(1)}\ln y_{\mathrm{Li}}^{(1)}\right) + y_{\mathrm{Al}}^{(1)}y_{\mathrm{Li}}^{(1)}\sum_v {}^vL_{\mathrm{Al,Li}:A}\left(x_{\mathrm{Al}}^{(1)} - x_{\mathrm{Li}}^{(1)}\right)^v \quad \text{(Eq 6)}$$

where $y_{\mathrm{Al}}^{(1)}$ and $y_{\mathrm{Li}}^{(1)}$ are the site fractions of the component of Al and Li on the first sublattice and A represents the Zr or Li component.

According to the experimental data measured by Zatorska et al.,[10] T2 has a homogeneity range at 470 K. According to the atomic parameters of $Al_3Li_{1.2}Zr_{4.8}$ and the formation process of T2 phase,[9] the four-sublattice model, $(\mathrm{Al,Zr})_{0\cdot375}(\mathrm{Li,Zr})_{0.025}(\mathrm{Zr,Li})_{0.1}\mathrm{Zr}_{0.5}$, is used to describe the T2 phase. Al, Li and Zr are major constituent for the first, second and third sublattice individually. Since the second sublattice occupies a small site fractions and Li is major constituent for this sublattice, this model is simplified to $(\mathrm{Al,Zr})_{0.375}\mathrm{Li}_{0.025}(\mathrm{Zr,Li})_{0.1}\mathrm{Zr}_{0.5}$. The expression of the Gibbs energy is as follow:

$$\begin{aligned} {}^0G_m^{T2} = &\sum_i\sum_j y_i^{(1)}y_j^{(3)}G_{i:\mathrm{Li}:j:\mathrm{Zr}} + \mathrm{RT}\left(0.375y_i^{(1)}\ln y_i^{(1)} + 0.1y_j^{(3)}\ln y_j^{(3)}\right) \\ &+ \sum_i\sum_{j>i}\sum_k\sum_v y_i^{(1)}y_j^{(1)}y_k^{(3)}{}^vL_{i,j:\mathrm{Li}:k:\mathrm{Zr}}^{T2}\left(y_i^{(1)} - y_j^{(1)}\right)^v \\ &+ \sum_i\sum_{j>i}\sum_k\sum_v y_i^{(3)}y_j^{(3)}y_k^{(1)}{}^vL_{k:\mathrm{Li}:i,j:\mathrm{Zr}}^{T2}\left(y_i^{(3)} - y_j^{(3)}\right)^v \\ &+ \sum_i\sum_{j>i}\sum_k\sum_{l>k} y_i^{(1)}y_j^{(1)}y_k^{(3)}y_l^{(3)}L_{i,j:\mathrm{Li}:k,l:\mathrm{Zr}} \end{aligned} \quad \text{(Eq 7)}$$

where $y_i^{(n)}$ (n = 1 or 3) is the site fractions of element in the 1st or 3rd sublattice.

3.4 Orderding Phase

In Al-Li binary system, the four-sublattice model was used to described ordered $L1_0$ (AlLi), $L1_2$ (Al_3Li and $AlLi_3$), B2 (AlLi), B32 (AlLi), $D0_3$ (Al_3Li and $AlLi_3$), disordered fcc(Al) and disordered bcc(Al) phases by Hallstedt and Kim.[17] The thermodynamic parameters determined by Hallstedt and Kim[17] were adopted in the present work.

4 First-Principle Calculations

In order to get a set of reasonable thermodynamic parameters of the Al-Li-Zr ternary system, the formation enthalpy of T1 has been calculated through the first-principle method.

The first-principle calculations base on the density functional theory (DFT).[69] In this work, the Vienna Ab-initio Simulation Package (VASP)[70] has been used to calculate the formation enthalpy of T1. The kinetic-energy cutoff was 520 eV for this calculation. The Brillouin-Zone (BZ) k-point sampling was performed using the Monkhorst-Pack scheme with a 5 × 5 × 5 mesh for the calculation of T1. The total energy was converged numerically to $< 1 \times 10^{-5}$ eV/unit with respect to electronic, ionic and unit cell degrees of freedom. The Hellmann–Feynman forces on each ion were < 0.01 eV/Å after geometer optimization.

The formation enthalpies for T1 were calculated by the equation as follow:

$$\Delta H^{\mathrm{Al}_x\mathrm{Li}_y\mathrm{Zr}_z} = E^{\mathrm{Al}_x\mathrm{Li}_y\mathrm{Zr}_z} - xE^{\mathrm{fcc(Al)}} - yE^{\mathrm{bcc(Li)}} - zE^{\mathrm{hcp(Zr)}} \quad \text{(Eq 8)}$$

where $E^{\mathrm{Al}_x\mathrm{Li}_y\mathrm{Zr}_z}$, $E^{\mathrm{fcc(Al)}}$, $E^{\mathrm{bcc(Li)}}$ and $E^{\mathrm{hcp(Zr)}}$ are the total energy of $\mathrm{Al}_x\mathrm{Li}_y\mathrm{Zr}_z$, pure Al, pure Li and pure Zr respectively.

5 Results and Discussion

In this work, the thermodynamic assessment of Al-Zr, Al-Li and Al-Li-Zr systems have been carried out using Thermo-Calc software package. All the thermodynamic parameters are given in Table 5.

5.1 Al-Zr Binary System

The calculated Al-Zr phase diagram from the present work is shown in Fig. 1(a) and (b) together with experimental data.[8,19–28] It shows that the calculated liquidus and solidus can reproduce the latest experimental data.[8] The detailed comparisons of temperatures and compositions for

Table 4 Enthalpies of formation of the stable Al-Zr and Al-Li system (kJ/mol)

Intermetallic compounds	This work	Ab-initio	References	Experiment	References	CALPHAD	References
Al-Zr binary system							
Al_3Zr	− 52.870	− 46.78	[38]	− 49.0 ± 4.0	[34]	− 40.600	[11]
		− 46.77	[39]	− 48.4 ± 1	[36]	− 40.500	[12]
		− 49.106	[40]	− 51.6 ± 1.2	[37]	− 44.298	[13]
		− 39.593	[41]			− 48.541	[14]
						− 46.779	[15]
						− 44.321	[16]
Al_2Zr	− 58.257	− 51.603	[38]	− 51.3 ± 4.3	[35]	− 45.800	[11]
		− 52.23	[39]	− 52.1 ± 1.6	[36]	− 45.810	[12]
		− 53.327	[40]	− 54.0 ± 4.0	[34]	− 45.536	[13]
		− 47.837	[41]	− 56.3 ± 2.2	[37]	− 52.582	[14]
						− 51.602	[15]
						− 51.446	[16]
Al_3Zr_2	− 57.664	− 49.867	[38]	55.0 ± 4.0	[34]	− 47.300	[11]
		− 49.84	[39]	− 59.6 ± 2.4	[37]	− 46.940	[12]
		− 51.649	[40]			− 45.200	[13]
		− 45.474	[41]			− 56.613	[14]
						− 49.899	[15]
						− 49.005	[16]
AlZr	− 55.400	− 44.875	[38]	− 53.0 ± 4.0	[34]	− 46.897	[11]
		− 44.79	[39]	− 71.3 ± 4.4	[37]	− 44.500	[12]
		− 46.163	[40]			− 44.685	[13]
		− 40.892	[41]			− 64.950	[14]
						− 45.326	[15]
						− 45.434	[16]
Al_4Zr_5	− 49.419	− 40.175	[38]	− 52.0 ± 4.0	[34]	− 45.597	[11]
		− 40.06	[39]			− 41.000	[12]
		− 42.002	[40]			− 43.127	[13]
		− 35.208	[41]			− 55.424	[14]
						− 38.725	[15]
						− 41.371	[16]
Al_3Zr_4	− 52.138	− 44.875	[38]			− 58.480	[14]
		− 46.21	[39]			− 40.894	[15]
		− 47.555	[40]			− 42.925	[16]
		− 39.000	[41]				
Al_2Zr_3	− 50.284	− 38.262	[38]	− 49.0 ± 4.0	[34]	− 42.670	[11]
		− 38.13	[39]	− 49.0 ± 3.1	[37]	− 38.427	[12]
		− 39.298	[40]			− 40.564	[13]
		− 36.463	[41]			− 55.180	[14]
						− 38.854	[15]
						41.929	[16]
Al_3Zr_5	− 47.803	− 35.308	[38]	− 48.0 ± 4.0	[34]	− 39.806	[11]
		− 36.02	[39]			− 36.248	[12]
		− 37.599	[40]			− 39.000	[13]
		− 33.283	[41]			− 51.484	[14]
						− 36.611	[15]
						− 39.973	[16]
$AlZr_2$	− 46.834	− 34.222	[38]	− 56.1 ± 3.1	[37]	− 36.250	[11]
		− 34.64	[39]			− 33.375	[12]

Table 4 continued

Intermetallic compounds	This work	Ab-initio	References	Experiment	References	CALPHAD	References
		− 36.753	[40]			− 39.145	[13]
		− 30.434	[41]			− 48.358	[14][14]
						− 35.979	[15]
						− 39.798	[16]
$AlZr_3$	− 37.448	− 28.994	[38]	− 43.6 ± 2.3	[37]	− 28.700	[11]
		− 29.15	[39]			− 27.000	[12]
		− 31.088	[40]			− 31.767	[13]
		− 23.117	[41]			− 36.163	[14]
						− 28.993	[15]
						− 37.283	[16]
Al-Li binary system							
AlLi (B32)	− 20.0	− 21.8	[63]	− 20.7	[64]	− 20.0	[17]
		− 20.3	[59]				
		− 21.9	[60]				
		− 23.8	[61]				
		− 20.7	[62]				
AlLi (B2)	− 14.0	6.1	[59]			− 14.0	[17]
		− 13.7	[60]				
		− 13.8	[62]				
$AlLi(L1_0)$	− 13.1	− 13.7	[60]			− 13.1	[17]
		− 11.9	[61]				
		− 9.5	[62]				
Al_2Li_3	− 18.8	− 19.0	[60]	− 19.1	[64]	− 18.8	[17]
		− 21.1	[62]				
$AlLi_2$	− 17.2						
Al_4Li_9	− 15.1	− 14.9	[60]	− 15.1	[64]	− 14.9	[17]
		− 16.8	[62]				
$Al_3Li(L1_2)$	− 8.1	− 11.1	[60]			− 8.1	[17]
		− 10.1	[61]				
		− 7.5	[62]				
$Al_3Li(D0_3)$	− 5.0	1.9	[60]			− 5.0	[17]
		− 3.9	[62]				
$AlLi_3(L1_2)$	− 5.0	− 6.5	[60]			− 5.0	[17]
		− 5.0	[61]				
		− 2.3	[62]				
$AlLi_3(D03)$	− 10.5	− 10.9	[60]			− 10.5	[17]
		− 10.2	[62]				

the invariant reactions between calculated and experimental data[8,19–23,28] are shown in Table 2. As can be seen, the calculated Al-Zr binary phase diagram agrees well with the reported experimental phase diagram.

Figure 2 and Table 4 show the calculated formation enthalpies of the Al-Zr binary compounds at 298 K with the literature data.[11–16,34–41] As shown in Fig. 2, the calculated enthalpies of formation are in reasonable agreement with most experimental results.

Figure 3 shows the calculated results of the mixing enthalpies of the liquid phase at 1790 K with the experimental data.[42–45] A good agreement has been achieved, especially from 0 to 25 at.% Zr.

Table 5 Thermodynamic parameters in Al-Li-Zr system

Phase	Thermodynamic parameters		References
Functions	for bcc(B2) and bcc(Li)		
	${}^{0}L^{4sl_b} = {}^{1}L^{4sl_b} = {}^{2}L^{4sl_b} = 0$		[17]
	$G^{4sl_b}_{Al_3Li} = 2u_1 + 1.5u_2 + \Delta G_{Al_3Li}$		[17]
	$G^{B2_b}_{Al_2Li_2} = 4u_1$	$G^{B32_b}_{Al_2Li_2} = 2u_1 + 3u_2$	[17]
	$G^{4sl_b}_{AlLi_3} = 2u_1 + 1.5u_2 + \Delta G_{AlLi_3}$		[17]
	$L^{4sl_b}_{rec} = 15000$		[17]
	$u_1 = -3360 + 1.8T$	$u_2 = -4320 + 1.86T$	[17]
	$\Delta G_{Al_3Li} = 3700$	$\Delta G_{Al_3Li} = 3250$	[17]
	for fcc(ord) and fcc(Al)		
	${}^{0}L^{4sl_f} = 2960 - 1.56T$	${}^{1}L^{4sl_f} = {}^{2}L^{4sl_f} = 0$	[17]
	$G^{4sl_f}_{Al_3Li} = 3u + \Delta G_{Al_3Li}$	$G^{4sl_f}_{Al_2Li_2} = 4u$	[17]
	$G^{4sl_f}_{AlLi_3} = 3u + \Delta G_{AlLi_3}$		[17]
	$L^{4sl_f}_{rec} = u$	$u = -3270 + 1.96T$	[17]
	$\Delta G_{Al_3Li} = 1750 - 4.7T$	$\Delta G_{Al_3Li} = 3250$	[17]
Liquid	Model $(Al,Li,Zr)_1$		
	${}^{0}L^{liq}_{Al,Li} = -44200 + 20.6T$	${}^{1}L^{liq}_{Al,Li} = 13600 - 5.3T$	[17]
	${}^{2}L^{liq}_{Al,Li} = 14200$	${}^{3}L^{liq}_{Al,Li} = -12100$	[17]
	${}^{4}L^{liq}_{Al,Li} = -7100$		[17]
	${}^{0}L^{liq}_{Li,Zr} = 80922.4 + 10.0357T$		[18]
	${}^{0}L^{liq}_{Al,Zr} = -186477.95 + 49.85850T$		This work
	${}^{1}L^{liq}_{Al,Zr} = -38386.64 + 25.91T$		This work
	${}^{2}L^{liq}_{Al,Zr} = 48183.91 - 7.04T$		This work
bcc(Li)	Model $(Al,Li)_1$		
	$${}^{0}L^{bcc(Li)}_{Al,Li} = G^{4sl_b}_{Al_3Li} + 0.5G^{B2_b}_{Al_2Li_2} + G^{B32_b}_{Al_2Li_2} + G^{4sl_b}_{AlLi_3} + 4\,{}^{0}L^{4sl_b}$$		[17]
	$${}^{1}L^{bcc(Li)}_{Al,Li} = 2G^{4sl_b}_{Al_3Li} - 2G^{4sl_b}_{AlLi_3} + 4\,{}^{1}L^{4sl_b}$$		[17]
	$${}^{2}L^{bcc(Li)}_{Al,Li} = G^{4sl}_{Al_3Li} - 0.5G^{B2}_{Al_2Li_2} - G^{B32}_{Al_2Li_2} + G^{4sl}_{AlLi_3} + 4\,{}^{2}L^{4sl_b}$$		[17]
bcc(Zr)	Model $(Al,Li,Zr)_1$		
	${}^{0}L^{bcc(Zr)}_{Li,Zr} = 85000$		[18]
	${}^{0}L^{bcc(Zr)}_{Al,Zr} = -154558.69 + 26.09T$		This work
	${}^{1}L^{bcc(Zr)}_{Al,Zr} = -27900$		This work
	${}^{2}L^{bcc(Zr)}_{Al,Zr} = 10000$		This work
bcc(B2)	Model $(Al,Li)_{0.25}(Al,Li)_{0.25}(Al,Li)_{0.25}(Al,Li)_{0.25}$		

Table 5 continued

Phase	Thermodynamic parameters	References
(AlLi)	${}^{0}G^{AlLi}_{Al:Al:Al:Al} = 0$; ${}^{0}G^{AlLi}_{Li:Li:Li:Li} = 0$	[17]
	${}^{0}G^{AlLi}_{Li:Al:Al:Al} = {}^{0}G^{AlLi}_{Al:Li:Al:Al} = {}^{0}G^{AlLi}_{Al:Al:Li:Al} = {}^{0}G^{AlLi}_{Al:Al:Al:Li} = G^{4sl_b}_{Al_3Li}$	[17]
	${}^{0}G^{AlLi}_{Li:Li:Al:Al} = {}^{0}G^{AlLi}_{Al:Al:Li:Li} = G^{B2_b}_{Al_2Li_2}$	[17]
	${}^{0}G^{AlLi}_{Li:Al:Li:Al} = {}^{0}G^{AlLi}_{Al:Li:Li:Al} = {}^{0}G^{AlLi}_{Al:Li:Al:Li} = {}^{0}G^{AlLi}_{Li:Al:Al:Li} = G^{B32_b}_{Al_2Li_2}$	[17]
	${}^{0}G^{AlLi}_{Al:Li:Li:Li} = {}^{0}G^{AlLi}_{Li:Al:Li:Li} = {}^{0}G^{AlLi}_{Li:Li:Al:Li} = {}^{0}G^{AlLi}_{Li:Li:Li:Al} = G^{4sl_b}_{AlLi_3}$	[17]
	${}^{0}G^{AlLi}_{Al,Li:Al,Li:*:*} = {}^{0}G^{AlLi}_{Al,Li:*:Al,Li:*} = {}^{0}G^{AlLi}_{Al,Li:*:*:Al,Li} = {}^{0}G^{AlLi}_{*:Al,Li:Al,Li:*} = {}^{0}G^{Al,Li}_{*:Al,Li:*:Al,Li} = {}^{0}G^{Al,Li}_{*:*:Al,Li:Al,Li} = L^{4sl_b}_{rec}$	
fcc(Al)	Model $(Al,Li,Zr)_1$	
	${}^{0}L^{fcc(Al)}_{Al,Li} = G^{4sl_f}_{Al_3Li} + 1.5G^{4sl_f}_{Al_2Li_2} + G^{4sl_f}_{AlLi_3} + 1.5L^{4sl_f}_{rec} + 4{}^{0}L^{4sl_f}$	[17]
	${}^{1}L^{fcc(Al)}_{Al,Li} = 2G^{4sl_f}_{Al_3Li} - 2G^{4sl_f}_{AlLi_3} + 4{}^{1}L^{4sl_f}$	[17]
	${}^{2}L^{fcc(Al)}_{Al,Li} = G^{4sl_f}_{Al_3Li} - 1.5G^{4sl_f}_{Al_2Li_2} + G^{4sl_f}_{AlLi_3} - 1.5L^{4sl_f}_{rec} + 4{}^{2}L^{4sl_f}$	[17]
	${}^{0}L^{fcc(Al)}_{Li,Zr} = 12000$	This work
	${}^{0}L^{fcc(Al)}_{Al,Zr} = -120947.00 + 15.26T$	This work
fcc(ord)	Model $(Al,Li)_{0.25}(Al,Li)_{0.25}(Al,Li)_{0.25}(Al,Li)_{0.25}$	
	${}^{0}G^{fcc(ord)}_{Al:Al:Al:Al} = 0$; ${}^{0}G^{fcc(ord)}_{Li:Li:Li:Li} = 0$	[17]
	${}^{0}G^{fcc(ord)}_{Li:Al:Al:Al} = {}^{0}G^{fcc(ord)}_{Al:Li:Al:Al} = {}^{0}G^{fcc(ord)}_{Al:Al:Li:Al} = {}^{0}G^{fcc(ord)}_{Al:Al:Al:Li} = G^{4sl_f}_{Al_3Li}$	[17]
	${}^{0}G^{fcc(ord)}_{Li:Al:Li:Al} = {}^{0}G^{fcc(ord)}_{Al:Li:Li:Al} = {}^{0}G^{fcc(ord)}_{Al:Li:Al:Li} = {}^{0}G^{fcc(ord)}_{Li:Al:Al:Li} = {}^{0}G^{fcc(ord)}_{Li:Li:Al:Al} = {}^{0}G^{fcc(ord)}_{Al:Al:Li:Li} = G^{4sl_f}_{Al_2Li_2}$	[17]
	${}^{0}G^{fcc(ord)}_{Al:Li:Li:Li} = {}^{0}G^{fcc(ord)}_{Li:Al:Li:Li} = {}^{0}G^{fcc(ord)}_{Li:Li:Al:Li} = {}^{0}G^{fcc(ord)}_{Li:Li:Li:Al} = G^{4sl_f}_{AlLi_3}$	[17]
	${}^{0}G^{fcc(ord)}_{Al,Li:Al,Li:*:*} = {}^{0}G^{fcc(ord)}_{Al,Li:*:Al,Li:*} = {}^{0}G^{fcc(ord)}_{Al,Li:*:*:Al,Li} = {}^{0}G^{fcc(ord)}_{*:Al,Li:Al,Li:*} = {}^{0}G^{fcc(ord)}_{*:Al,Li:*:Al,Li} = {}^{0}G^{fcc(ord)}_{*:*:Al,Li:Al,Li} = L^{4sl_f}_{rec}$	[17]

Table 5 continued

Phase	Thermodynamic parameters	References
hcp(Zr)	Model $(Al,Li,Zr)_1$	
	${}^{0}L^{hcp(Zr)}_{Li,Zr} = 85000$	[17]
	${}^{0}L^{hcp(Zr)}_{Al,Zr} = -155043.50 + 29.43T$	This work
	${}^{1}L^{hcp(Zr)}_{Al,Zr} = 10002$	This work
	${}^{2}L^{hcp(Zr)}_{Al,Zr} = 25000$	This work
Al_2Li_3	Model $(Al)_{0.4}(Li)_{0.6}$	
	$G^{Al_2Li_3}_{Al:Li} = 0.4GHSERAL + 0.6GHSERLI - 18798 + 6.9T$	[17]
Al_4Li_9	Model $(Al)_{0.3077}(Li)_{0.6923}$	
	$G^{Al_4Li_9}_{Al:Li} = 0.3077GHSERAL + 0.6923GHSERLI - 15111.4 + 6.01T$	This work
$AlLi_2$	Model $(Al,Li)_{0.36}(Li)_{0.64}$	
	$G^{AlLi_2}_{Li:Li} = GHSRELI + 10000$	This work
	$G^{AlLi_2}_{Al:Li} = 0.36GHSREAL + 0.64GHSERLI - 17192 + 6.54T$	This work
	${}^{0}L^{AlLi_2}_{Al,Li:Li} = -17880 + 19.2T$	This work
	${}^{1}L^{AlLi_2}_{Al,Li:Li} = 4440 - 6T$	This work
$AlZr_3$	Model $(Al)_{0.25}(Zr)_{0.75}$	
	$G^{AlZr_3}_{Al:Zr} = 0.25GHESEAL + 0.75GHSERZR - 37448.0 + 8.423T$	This work
$AlZr_2$	Model $(Al)_{0.33333}(Zr)_{0.66667}$	
	$G^{AlZr_2}_{Al:Zr} = 0.33333GHESEAL + 0.66667GHSERZR - 46834.2 + 9.382T$	This work
Al_3Zr_5	Model $(Al)_{0.375}(Zr)_{0.625}$	
	$G^{Al_3Zr_5}_{Al:Zr} = 0.375GHESEAL + 0.625GHSERZR - 47803.0 + 7.675T$	This work
Al_2Zr_3	Model $(Al,Li)_{0.4}(Zr)_{0.6}$	
	$G^{Al_2Zr_3}_{Al:Zr} = 0.4GHESEAL + 0.6GHSERZR - 50284.0 + 8.139T$	This work
	${}^{0}L^{Al_2Zr_3}_{Al:Zr} = -35000$	This work
	${}^{1}L^{Al_2Zr_3}_{Al:Zr} = -32000$	This work
Al_3Zr_4	Model $(Al,Li)_{0.42857}(Zr)_{0.57143}$	
	$G^{Zr_4Al_3}_{Al:Zr} = 0.42857GHESEAL + 0.57143GHSERZR - 52138.1 + 8.895T$	This work
	${}^{0}L^{Al_3Zr_4}_{Al:Zr} = -37600$	This work
	${}^{1}L^{Al_3Zr_4}_{Al:Zr} = -33500$	This work
Al_4Zr_5	Model $(Al)_{0.44444}(Zr)_{0.55556}$	
	$G^{Al_4Zr_5}_{Al:Zr} = 0.44444GHESEAL + 0.55556GHSERZR - 49418.8 + 6.427T$	This work

Table 5 continued

Phase	Thermodynamic parameters	References
AlZr	Model $(Al,Li)_{0.5}(Zr)_{0.5}$	
	$G^{AlZr}_{Al:Zr} = 0.5GHESEAL + 0.5GHSERZR - 55400.3 + 9.919T$	This work
	${}^{0}L^{AlZr}_{Al,Li:Zr} = -46550$	This work
	${}^{1}L^{AlZr}_{Al,Li:Zr} = -41000$	This work
Al_3Zr_2	Model $(Al,Li)_{0.6}(Zr)_{0.4}$	
	$G^{Al_3Zr_2}_{Al:Zr} = 0.6GHESEAL + 0.4GHSERZR - 57644.8 + 10.682T$	This work
	${}^{0}L^{Al_3Zr_2}_{Al,Li:Zr} = -74500$	This work
	${}^{1}L^{Al_3Zr_2}_{Al,Li:Zr} = 2000$	This work
	${}^{2}L^{Al_3Zr_2}_{Al,Li:Zr} = -53800$	This work
Al_2Zr	Model $(Al,Li)_{0.66667}(Zr)_{0.33333}$	
	$G^{Al_2Zr}_{Al:Zr} = 0.66667GHESEAL + 0.33333GHSERZR - 58257.2 + 11.626T$	This work
	${}^{0}L^{Al_2Zr}_{Al:Zr} = -37500$	This work
	${}^{1}L^{Al_2Zr}_{Al:Zr} = -38500$	This work
	${}^{2}L^{Al_2Zr}_{Al:Zr} = -48500$	This work
Al_3Zr	Model $(Al,Li)_{0.75}(Zr)_{0.25}$	
	$G^{Al_3Zr}_{Al:Zr} = 0.75GHESEAL + 0.25GHSERZR - 52870.3 + 11.573T$	This work
	${}^{0}L^{Al_3Zr}_{Al:Zr} = -5500$	This work
	${}^{0}L^{Al_3Zr}_{Al:Zr} = -38000$	This work
	${}^{0}L^{Al_3Zr}_{Al:Zr} = -52000$	This work
T1	Model $(Al)_{0.25}(Li)_{0.5}(Zr)_{0.25}$	
	$G^{T1}_{Al:Li:Zr} = 0.25GHESEAL + 0.5GHSERLi + 0.25GHSERZR - 3894.6 - 65.88 * T$	This work
T2	Model $(Al,Zr)_{0.375}(Li)_{0.025}(Zr,Li)_{0.1}(Zr)_{0.5}$	
	$G^{T2}_{Al:Li:Li:Zr} = 0.375GHSERAL + 0.125GHSERLI + 0.5GHESEZR - 51280$	This work
	$G^{T2}_{Al:Zr:Li:Zr} = 0.375GHSERAL + 0.025GHSERLI + 0.6GHESEZR - 46660$	This work
	${}^{0}L^{T2}_{Al:Li:Zr,Li:Zr} = -1800$; ${}^{1}L^{T2}_{Al:Li:Zr,Li:Zr} = -1550$	This work
	${}^{0}L^{T2}_{Al,Zr:Li:Li:Zr} = -15500$; ${}^{1}L^{T2}_{Al,Zr:Li:Li:Zr} = -8000$	This work
	${}^{0}L^{T2}_{Al,Zr:Li:Zr:Zr} = 7400$	This work
	${}^{0}L^{T2}_{Al,Zr:Li:Zr,Li:Zr} = -43000$	This work

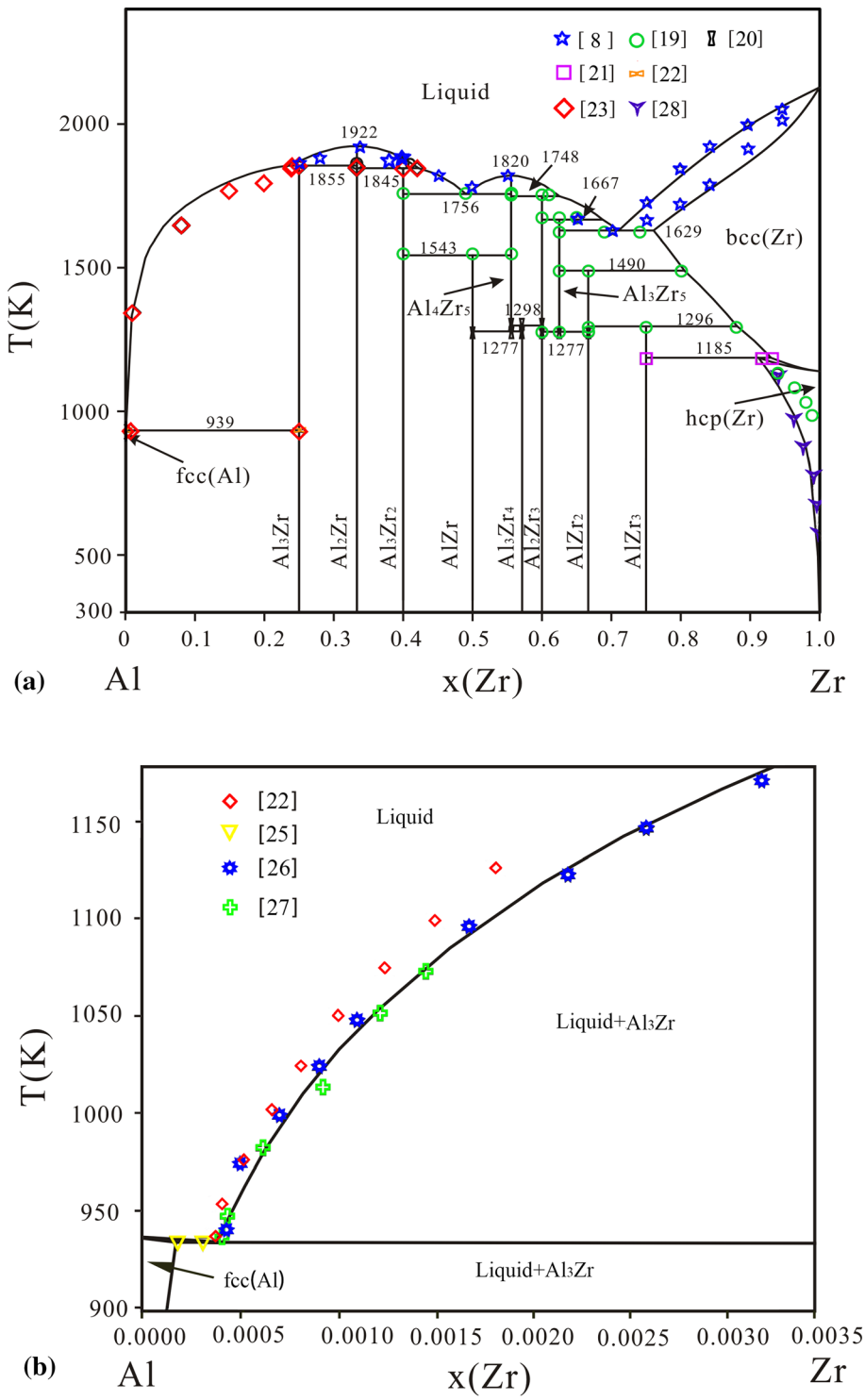

Fig. 1 (a) Calculated Al-Zr binary phase diagram, compared with the experimental data.[8,19–23, 28] (b) Enlarged Al-rich part of the optimized Al-Zr phase diagram[22,25–27]

5.2 Al-Li Binary System

According to a recent report,[9] there is a compound $AlLi_2$ in the Al-Li binary phase diagram. Therefore, the Al-Li binary system has been reassessed combing with the most of thermodynamic parameters reported by Hallstedt and Kim[17] in the present work.

The calculated phase diagram of Al-Li binary system is shown in Fig. 4 and the comparison of invariant equilibria data is listed in Table 3. For one thing, most of phase equilibrium information is consistent with the calculated

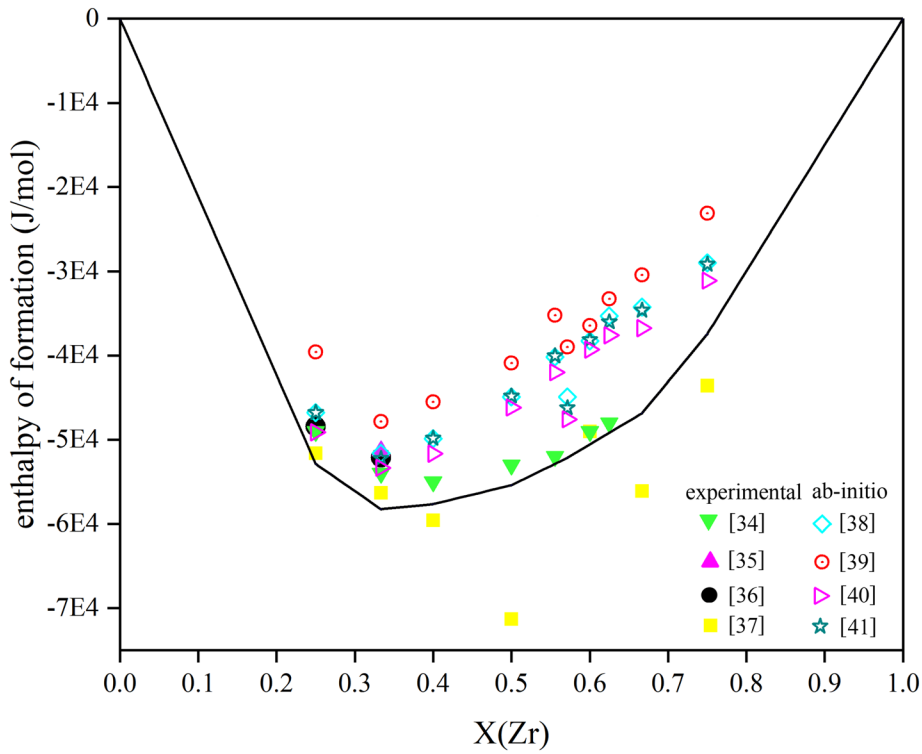

Fig. 2 Formation enthalpies of the stable phase in Al-Zr binary system from the present calculation, ab-initio calculations[38–41] and experimental data[34–37]

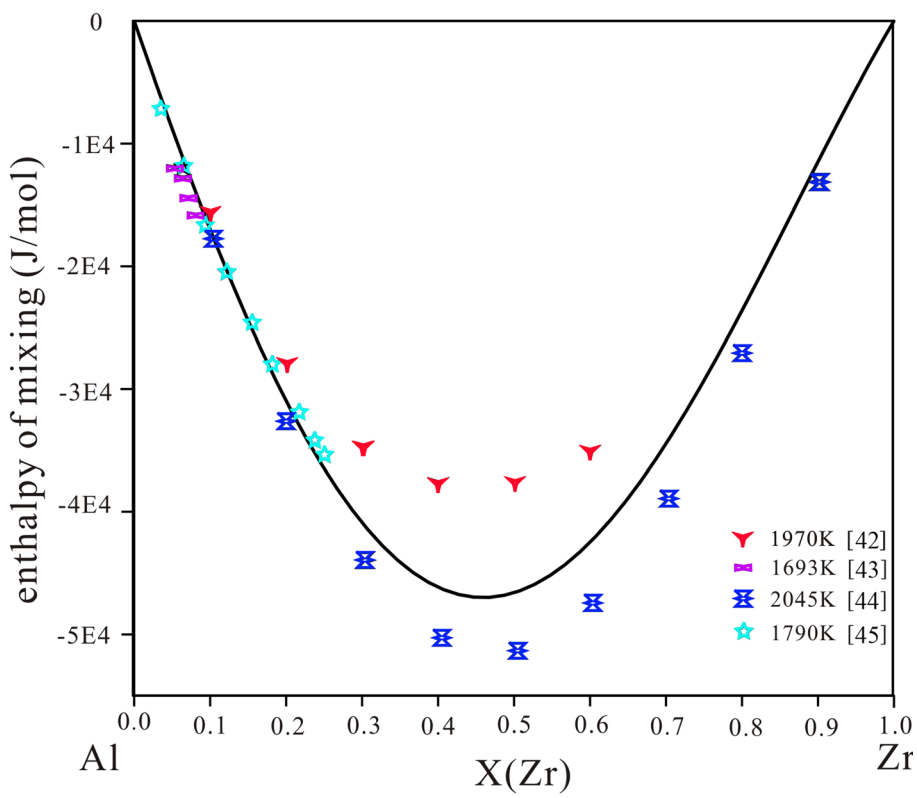

Fig. 3 Calculated enthalpy of mixing for the liquid phase at 1790 K and compared to the experimental values[42–45]

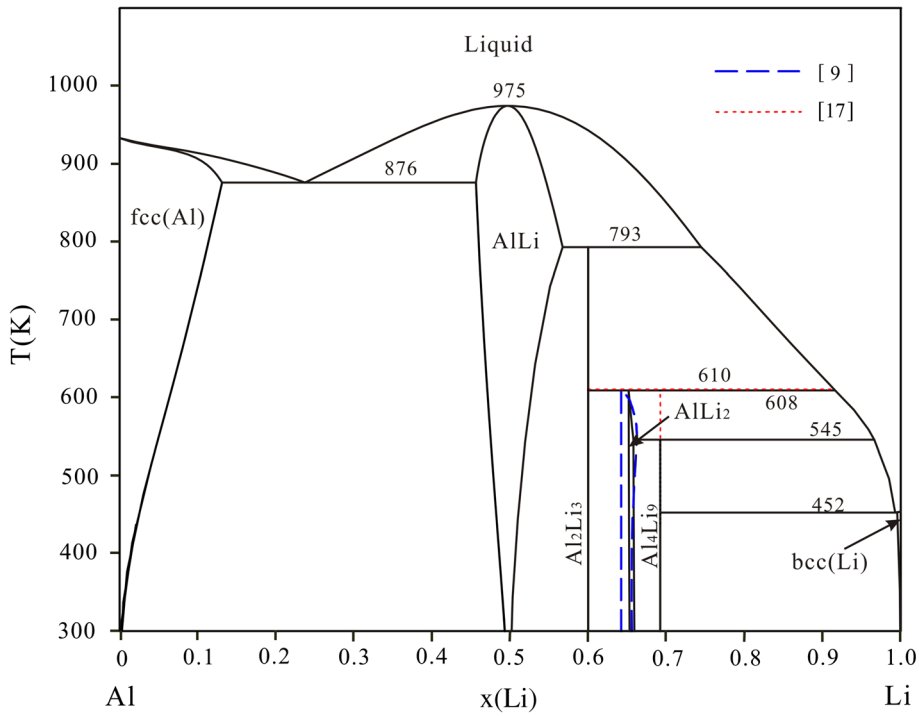

Fig. 4 Calculated Al-Li binary phase diagram, compared with the experimental data[9] and calculated data[17]

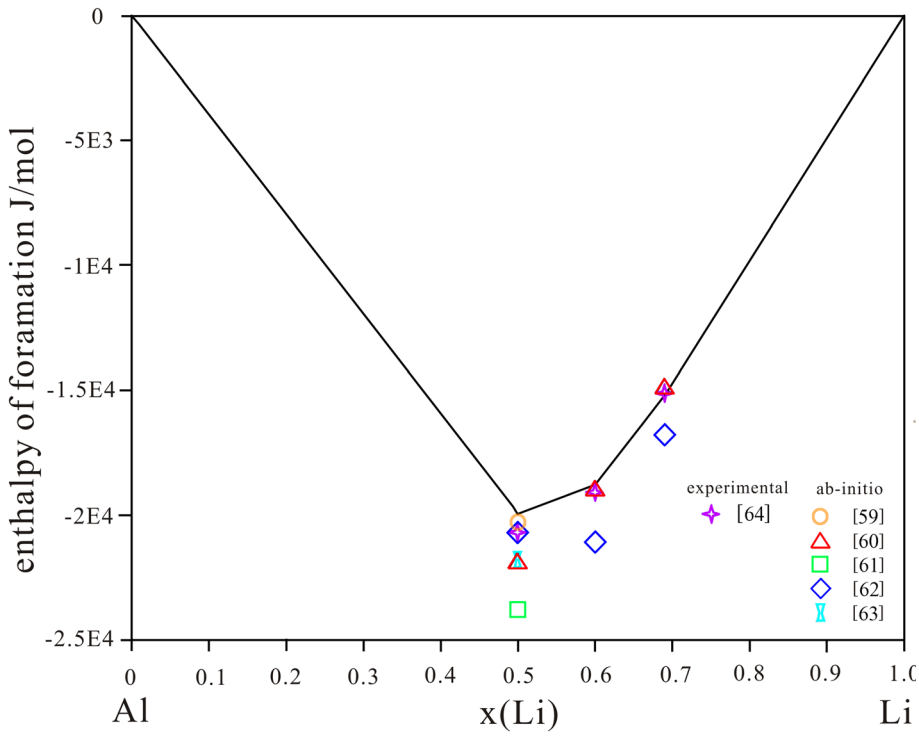

Fig. 5 Formation enthalpies of the stable phase in Al-Li binary system from the present calculation, ab-initio calculations[64] and experimental data[59–63]

Table 6 The calculated formation enthalpy of the Al-Li-Zr ternary compounds

Phase	Space group	Lattice parameter, Å a	First-principle (ΔH, kJ/mol)	CALPHAD (ΔH, kJ/mol)	References
$AlLi_2Zr$	$F\bar{4}3m$	6.598 6.633	− 3.4662	− 3.8946	This work [10]

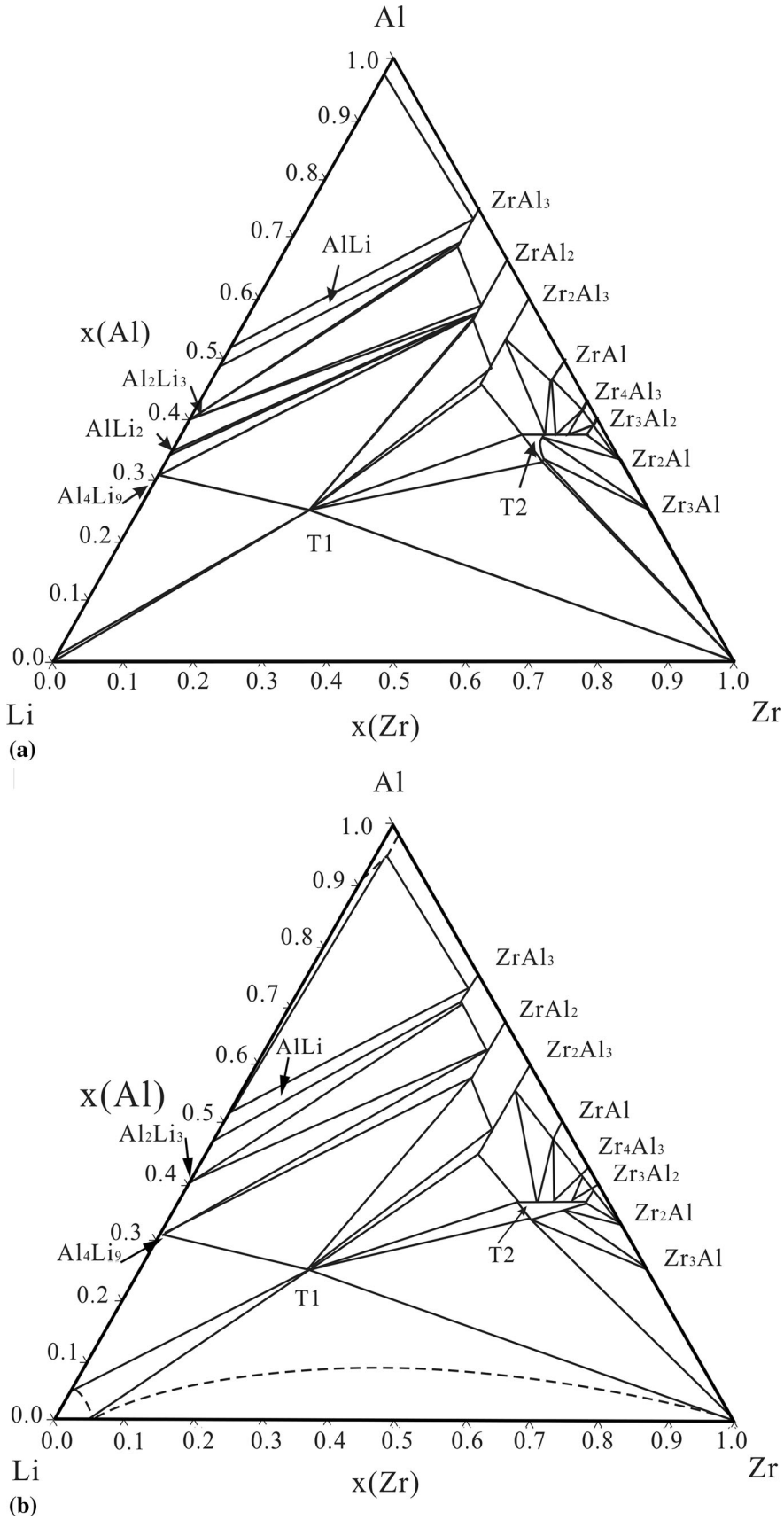

Fig. 6 Calculated isothermal section of the Al-Li-Zr system at 470 K, compared with the experimental data:[10] (a) calculated and (b) experimental

phase diagram.[17] For another, the phase relations between $AlLi_2$ and other phases are in good agreement with the latest experiment data.[9]

The calculated enthalpies of formation in Al-Li binary system are compared with the literature results[59–64] in Fig. 5 and Table 4. The formation enthalpies of Al_4Li_9, AlLi and Al_2Li_3 are in good agreement with the experimental data.[64] As shown in Fig. 5, the calculated formation enthalpy of $AlLi_2$ is also rational.

5.3 Al-Li-Zr Ternary System

In the present work, two ternary compounds, i.e. T1 and T2, have been considered and the parameters of them have been assessed. Furthermore, the calculated formation enthalpy of T1 (listed in Table 6) obtained in this work has been adopted. The calculated 470 K isothermal section of Al-Li-Zr ternary system is shown in Fig. 6(a) and the experimental isothermal section measured by Zatorska et al.[10] is shown in Fig. 6(b). It is worth pointing out that the thermodynamic description obtained in this work can reproduce most experimental phase relations[10] including T1 and T2. One two-phase equilibria $AlLi_2 + Al_2Zr$ and two three-phase equilibria $Al_4Li_9 + AlLi_2 + Al_2Zr$ and $Al_2Li_3 + AlLi_2 + Al_2Zr$ were presented in this work, which is attributed to the inclusion of new phase $AlLi_2$ in the thermodynamic assessment. As can be seen from Fig. 6(a) and (b), the calculated solubility of Li in AlZr, Al_3Zr_4 and Al_2Zr_3 are in reasonable agreement with the experimental results.[10] The calculated homogeneity region of the ternary compound (T2) can extend from 50 to 60 at.% Zr and 32 to 37 at.% Al at 470 K, which was close to the experimental results.[10] The difference between the calculated and determined isothermal section at 470 K is the phase boundary of T2. According to the experimental data, the boundary of T2 phase region is a triangle. However, one of the three sides is a curve in this work.

In the process of optimization, only one isothermal section of Al-Li-Zr ternary system can be referred and it is difficult to calculate the formation enthalpy of T2 through first-principle method, so the assessed parameters of T2 are unrelated to temperature. This results in the liquidus projection unreliable. Therefore, the liquidus projection is not presented in this work.

6 Conclusions

Based on the recent experimental phase relations and the calculated formation enthalpy of ternary compound T1, Al-Zr, Al-Li, and Al-Li-Zr systems have been assessed using CALPHAD method. The agreement between the calculated and experimental phase diagram was satisfactory.

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Data Availability The raw data and the processed data required to reproduce these findings are available from the corresponding author upon request, but most of them are included within the article.

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